

Johnson C. Smith University

Department of Natural Science and Mathematics

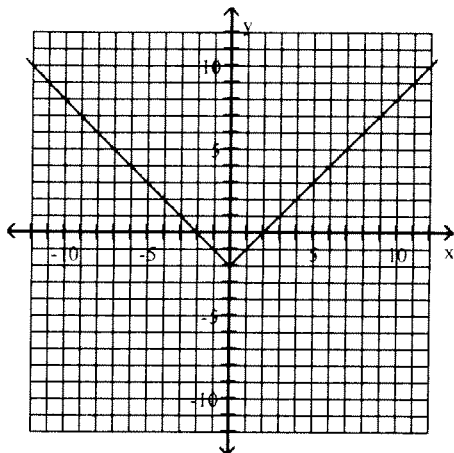
COMMON MIDTERM REVIEW - MTH-131 - COLLEGE ALGEBRA

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

Use the graph to determine the x- and y-intercepts.

1)

1) _____



A) x-intercepts: -2, 2; y-intercept: -2

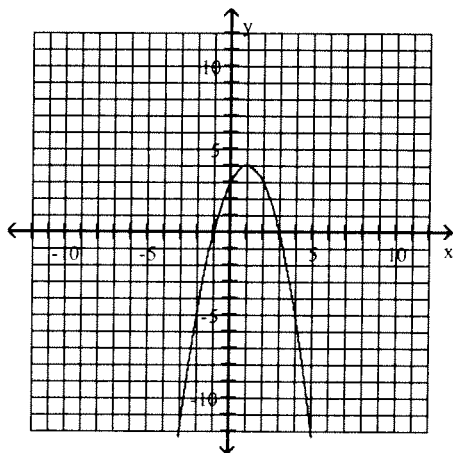
C) x-intercepts: -2, 2; y-intercept: 0

B) y-intercept: -2

D) x-intercepts: -2, 2

2)

2) _____



A) x-intercept: 3; y-intercepts: -1, 3

C) x-intercept: -1; y-intercepts: 3, 3

B) x-intercepts: -1, 3; y-intercept: 3

D) x-intercept: 3; y-intercept: 3

Solve and check the linear equation.

3) $7x - 9 = -10 + 4x$

3) _____

A) $\{3\}$

B) $\left\{-\frac{11}{19}\right\}$

C) $\left\{-\frac{1}{3}\right\}$

D) $\{-3\}$

4) $-3x + 5 + 6(x + 1) = -2x + 3$

A) $\left\{\frac{8}{3}\right\}$

B) $\left\{\frac{2}{5}\right\}$

C) $\left\{-\frac{8}{5}\right\}$

D) $\left\{-\frac{2}{3}\right\}$

4) _____

Determine whether the equation is an identity, a conditional equation, or an inconsistent equation.

5) $5x + 4x = 8x$

A) Identity

B) Conditional equation

C) Inconsistent equation

5) _____

6) $4(x + 1) + 54 = 9x - 5(x - 8)$

A) Identity

B) Conditional equation

C) Inconsistent equation

6) _____

Solve the problem.

7) A car rental agency charges \$250 per week plus \$0.10 per mile to rent a car. The total cost, C, for the renting the car for one week and driving it x miles can be modeled by the formula $C = 0.10x + 250$.

How many miles can you travel in one week for \$330?

A) 283 miles

B) 3300 miles

C) 800 miles

D) 775 miles

7) _____

Use the five-step strategy for solving word problems to find the number or numbers described in the following exercise.

8) When four times the number is added to 9 times the number, the result is 65. What is the number?

A) -7.2

B) 7.2

C) 0.6

D) 5

8) _____

9) When a number is decreased by 25% of itself, the result is 216. What is the number?

A) 288

B) 72

C) 1152

D) 9

9) _____

Solve the formula for the specified variable.

10) $A = \frac{1}{2}bh$ for b

A) $b = \frac{Ah}{2}$

B) $b = \frac{A}{2h}$

C) $b = \frac{h}{2A}$

D) $b = \frac{2A}{h}$

10) _____

11) $F = \frac{9}{5}C + 32$ for C

A) $C = \frac{5}{9}(F - 32)$

B) $C = \frac{5}{F - 32}$

C) $C = \frac{F - 32}{9}$

D) $C = \frac{9}{5}(F - 32)$

11) _____

12) $P = 2L + 2W$ for W

A) $W = \frac{P - 2L}{2}$

B) $W = P - L$

C) $W = P - 2L$

D) $W = \frac{P - L}{2}$

12) _____

Solve the equation by factoring.

13) $x^2 = x + 6$

A) {1, 6}

B) {2, 3}

C) {-2, 3}

D) {-2, -3}

13) _____

14) $2x^2 - 15x = 8$

A) {-2, 8}

B) $\left\{\frac{1}{15}, -\frac{1}{2}\right\}$

C) $\left\{-\frac{1}{2}, 8\right\}$

D) $\left\{-\frac{1}{2}, 2\right\}$

14) _____

Solve the equation by the square root property.

15) $(x - 3)^2 = 49$

A) $\{52\}$

B) $\{-7, 7\}$

C) $\{-4, 10\}$

D) $\{-10, -4\}$

15) _____

16) $2(x - 6)^2 = 10$

A) $\{6 \pm \sqrt{5}\}$

B) $\{-11, -1\}$

C) $\{-6 \pm \sqrt{5}\}$

D) $\{1, 11\}$

16) _____

Solve the equation using the quadratic formula.

17) $x^2 + 7x + 7 = 0$

A) $\left\{ \frac{-7 - \sqrt{21}}{2}, \frac{-7 + \sqrt{21}}{2} \right\}$

B) $\left\{ \frac{-7 - \sqrt{77}}{2}, \frac{-7 + \sqrt{77}}{2} \right\}$

C) $\left\{ \frac{-7 - \sqrt{21}}{14}, \frac{-7 + \sqrt{21}}{14} \right\}$

D) $\left\{ \frac{7 - \sqrt{21}}{2}, \frac{7 + \sqrt{21}}{2} \right\}$

17) _____

18) $6x^2 = -12x - 4$

A) $\left\{ \frac{-3 - \sqrt{3}}{12}, \frac{-3 + \sqrt{3}}{12} \right\}$

B) $\left\{ \frac{-3 - \sqrt{15}}{3}, \frac{-3 + \sqrt{15}}{3} \right\}$

C) $\left\{ \frac{-3 - \sqrt{3}}{3}, \frac{-3 + \sqrt{3}}{3} \right\}$

D) $\left\{ \frac{-12 - \sqrt{3}}{3}, \frac{-12 + \sqrt{3}}{3} \right\}$

18) _____

Solve the problem.

19) A ladder that is 17 feet long is 8 feet from the base of a wall. How far up the wall does the ladder reach?

A) 3 ft

B) $\sqrt{353}$ ft

C) 225 ft

D) 15 ft

19) _____

20) The length of a rectangular storage room is 2 feet longer than its width. If the area of the room is 195 square feet, find its dimensions.

A) 14 feet by 16 feet

B) 12 feet by 16 feet

C) 13 feet by 15 feet

D) 12 feet by 14 feet

20) _____

Solve the polynomial equation by factoring and then using the zero product principle.

21) $x^3 - 2x^2 - 24x = 0$

A) $\{0, -4, 6\}$

B) $\{-4, 6\}$

C) $\{4, -6\}$

D) $\{0, 4, -6\}$

21) _____

22) $x^3 + 2x^2 - x - 2 = 0$

A) $\{-2, 2\}$

B) $\{1, -2, 2\}$

C) $\{4\}$

D) $\{-1, 1, -2\}$

22) _____

Solve the radical equation, and check all proposed solutions.

23) $\sqrt{x + 1} = 3$

A) $\{9\}$

B) $\{8\}$

C) $\{10\}$

D) $\{16\}$

23) _____

24) $\sqrt{5x + 24} = x$

A) $\{-3, 8\}$

B) $\{8\}$

C) $\emptyset$

D) $\{-6\}$

24) _____

Solve the absolute value equation or indicate that the equation has no solution.

25) $|x + 3| = 3$

A) $\{0\}$

B) $\{-6, 0\}$

C) $\{0, 6\}$

D) $\emptyset$

25) _____

- 26) $3|x - 3| = 18$ _____
 A) $\{3, -9\}$ B) $\{9, -3\}$ C) $\{3\}$ D) $\emptyset$
- 27) $|6x - 4| - 3 = -12$ _____
 A) $\left\{-\frac{5}{6}, -\frac{13}{6}\right\}$ B) $\left\{\frac{13}{6}, \frac{5}{6}\right\}$ C) $\left\{-\frac{5}{6}\right\}$ D) $\emptyset$

Give the domain and range of the relation.

- 28) $\{(3, -9), (-6, -5), (1, 3), (1, 8)\}$ _____
 A) domain = $\{1, 3, -6, -1\}$; range = $\{3, -9, -5, 8\}$
 B) domain = $\{3, -9, -5, 8\}$; range = $\{1, 3, -6\}$
 C) domain = $\{1, 3, -6\}$; range = $\{3, -9, -5, 8\}$
 D) domain = $\{1, 3, -6, 11\}$; range = $\{3, -9, -5, 8\}$
- 29) $\{(1, -2), (-3, -1), (-3, 0), (-2, 1), (6, 3)\}$ _____
 A) domain: $\{-2, -1, 1, 3\}$; range: $\{1, -2, -3, 6\}$
 B) domain: $\{1, -2, -3, 6\}$; range: $\{-2, -1, 1, 3\}$
 C) domain: $\{-2, -1, 0, 1, 3\}$; range: $\{1, -2, -3, 6\}$
 D) domain: $\{1, -2, -3, 6\}$; range: $\{-2, -1, 0, 1, 3\}$

Determine whether the relation is a function.

- 30) $\{(-2, -7), (3, -5), (6, -4), (9, -6), (10, -1)\}$ _____
 A) Function B) Not a function
- 31) $\{(-7, -1), (-7, 2), (-1, 8), (3, 3), (10, -7)\}$ _____
 A) Function B) Not a function
- 32) $\{(-9, 1), (-9, 9), (1, -5), (6, -6), (9, -5)\}$ _____
 A) Function B) Not a function

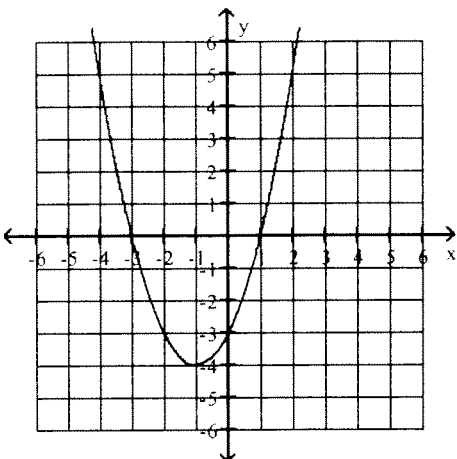
Evaluate the function at the given value of the independent variable and simplify.

- 33) $f(x) = -3x - 8$; $f(-2)$ _____
 A) -2 B) 22 C) -11 D) 14
- 34) $f(x) = \sqrt{x + 11}$; $f(-2)$ _____
 A) 1.73 B) -3
 C) 3 D) not a real number
- 35) $g(x) = 3x + 2$; $g(x - 1)$ _____
 A) $\frac{1}{3}x + 2$ B) $3x - 1$ C) $3x + 1$ D) $3x + 2$

Use the graph to determine the function's domain and range.

36)

36) _____



A) domain: $(-\infty, \infty)$

range: $[-4, \infty)$

C) domain: $[-1, \infty)$

range: $[-4, \infty)$

B) domain: $(-\infty, \infty)$

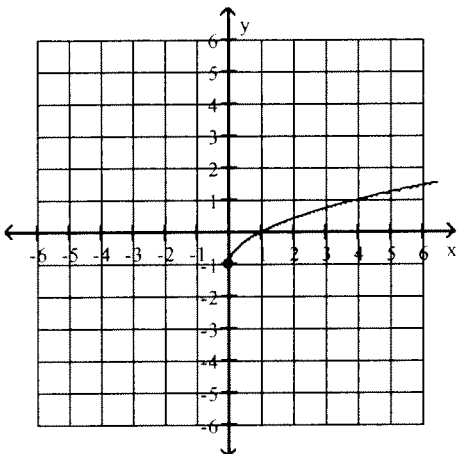
range: $(-\infty, \infty)$

D) domain: $(-\infty, -1)$ or $(-1, \infty)$

range: $(-\infty, -4)$ or $(-4, \infty)$

37)

37) _____



A) domain: $[0, \infty)$

range: $(-\infty, \infty)$

B) domain: $[0, \infty)$

range: $[0, \infty)$

C) domain: $(-\infty, \infty)$

range: $[-1, \infty)$

D) domain: $[0, \infty)$

range: $[-1, \infty)$

Determine whether the given function is even, odd, or neither.

38) $f(x) = x^3 - 4x$

A) Even

B) Neither

C) Odd

38) _____

39) $f(x) = 5x^2 + x^4$

A) Odd

B) Neither

C) Even

39) _____

40) $f(x) = x^4 - x^3$

A) Odd

B) Even

C) Neither

40) _____

41) $f(x) = x^3 + x^2 + 4$

A) Even

B) Neither

C) Odd

41) _____

Find the slope of the line that goes through the given points.

42) $(3, -5), (-9, -8)$

A) $-\frac{1}{4}$

B) $\frac{1}{4}$

C) $\frac{13}{6}$

D) 4

42) _____

43) $(-8, 8), (-5, 2)$

A) $-\frac{1}{2}$

B) $\frac{16}{7}$

C) $\frac{7}{16}$

D) -2

43) _____

Use the given conditions to write an equation for the line in point-slope form.

44) Slope = -4, passing through (7, 4)

A) $x - 4 = -4(y - 7)$

B) $y = -4x + 32$

C) $y + 4 = -4(x + 7)$

D) $y - 4 = -4(x - 7)$

44) _____

Use the given conditions to write an equation for the line in slope-intercept form.

45) Slope = -4, passing through (8, 4)

A) $y - 4 = -4x - 8$

B) $y - 4 = x - 8$

C) $y = -4x + 36$

D) $y = -4x - 36$

45) _____

46) Slope = $\frac{2}{3}$, passing through (7, 3)

A) $y = \frac{2}{3}x + 7$

B) $y = mx - \frac{5}{3}$

C) $y = \frac{2}{3}x - \frac{5}{3}$

D) $y = \frac{2}{3}x + \frac{5}{3}$

46) _____

Solve the equation.

47) $\frac{2x}{5} = \frac{x}{3} + 5$

A) $\{-150\}$

B) $\{150\}$

C) $\{75\}$

D) $\{-75\}$

47) _____

48) $\frac{7x}{3} - x = \frac{x}{27} - \frac{8}{9}$

A) $\left\{\frac{24}{35}\right\}$

B) $\left\{-\frac{24}{37}\right\}$

C) $\left\{-\frac{24}{35}\right\}$

D) $\left\{\frac{24}{37}\right\}$

48) _____

49) $\frac{x+18}{6} + \frac{x+9}{9} = x - 3$

A) $\left\{\frac{18}{13}\right\}$

B) $\left\{\frac{90}{13}\right\}$

C) $\left\{\frac{126}{13}\right\}$

D) $\left\{-\frac{18}{13}\right\}$

49) _____

First, write the value(s) that make the denominator(s) zero. Then solve the equation.

50) $\frac{7}{x} = \frac{1}{2x} + 26$

A) $x \neq 0, 2; \left\{\frac{15}{26}\right\}$

B) $x \neq 0; \left\{\frac{1}{4}\right\}$

C) No restrictions; $\{2\}$

D) $x \neq 0; \{4\}$

50) _____

Answer Key

Testname: REVIEW COMMON MIDTERM

- 1) A
- 2) B
- 3) C
- 4) C
- 5) B
- 6) C
- 7) C
- 8) D
- 9) A
- 10) D
- 11) A
- 12) A
- 13) C
- 14) C
- 15) C
- 16) A
- 17) A
- 18) C
- 19) D
- 20) C
- 21) A
- 22) D
- 23) B
- 24) B
- 25) B
- 26) B
- 27) D
- 28) C
- 29) D
- 30) A
- 31) B
- 32) B
- 33) A
- 34) C
- 35) B
- 36) A
- 37) D
- 38) C
- 39) C
- 40) C
- 41) B
- 42) B
- 43) D
- 44) D
- 45) C
- 46) C
- 47) C
- 48) C
- 49) C
- 50) B